tarTarea 10

R1

A	X	B	Y
7	2	6	11
3	4	9	15
10	7	2	4
1	12	2	11

R2

B	W	D	Y	A	Z
2	5	6	11	1	30
4	7	8	4	7	8
9	10	11	28	5	12

Tabla que incluya:

- *atributos*
- *registros*

$$R1 X R2$$

$$R2 \bowtie R1$$

$$R1 \bowtie_{((R1.A > R2.Z \text{ or } R1.A >= R2.W) \text{ and } R1.Y = R2.Y)} R2$$

Producto Cartesiano

R1 x R2

A	x	B	y	D	w	D	y	A	z
7	2	6	11	2	5	6	11	1	30
7	2	6	11	4	7	8	4	7	8
7	2	6	11	9	10	11	28	5	12
3	4	9	15	2	5	6	11	1	30
3	4	9	5	4	7	8	4	7	8
3	4	9	5	9	10	11	28	5	12
10	7	2	4	2	5	6	11	1	30
10	7	2	4	4	7	8	4	7	8
10	7	2	4	9	10	11	28	5	12
1	12	2	11	2	5	6	11	1	30
1	12	2	11	4	7	8	4	7	8
1	12	2	11	9	10	11	28	5	12

Cross Join

Creación de tablas

```sql
CREATE TABLE r1(
a int,
x int,
b int,
y int);

CREATE TABLE r2(
b int,
w int,
d int,
y int,
a int,
z int);

INSERT INTO r1(a, x, b, y) VALUES
(7, 2, 6, 11),
(3, 4, 9, 15),
(10, 7, 2, 4),
(1, 12, 2, 11);

INSERT INTO r2(b, w, d, y, a, z) VALUES
(2, 5, 6, 11, 1, 30),
(4, 7, 8, 4, 7, 8),
(9, 10, 11, 28, 5, 12);
```

```sql
26 SELECT * FROM r1;
```

Data Output | Messages | Notifications

SQL

	a integer	x integer	b integer	y integer
1	7	2	6	11
2	3	4	9	15
3	10	7	2	4
4	1	12	2	11

```sql
27 SELECT * FROM r2;
```

Data Output | Messages | Notifications

SQL

	b integer	w integer	d integer	y integer	a integer	z integer
1	2	5	6	11	1	30
2	4	7	8	4	7	8
3	9	10	11	28	5	12

CROSS JOIN

```sql
28 SELECT * FROM r1 CROSS JOIN r2;
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 12

	a integer	x integer	b integer	y integer	b integer	w integer	d integer	y integer	a integer	z integer
1	7	2	6	11	2	5	6	11	1	30
2	7	2	6	11	4	7	8	4	7	8
3	7	2	6	11	9	10	11	28	5	12
4	3	4	9	15	2	5	6	11	1	30
5	3	4	9	15	4	7	8	4	7	8
6	3	4	9	15	9	10	11	28	5	12
7	10	7	2	4	2	5	6	11	1	30
8	10	7	2	4	4	7	8	4	7	8
9	10	7	2	4	9	10	11	28	5	12
10	1	12	2	11	2	5	6	11	1	30
11	1	12	2	11	4	7	8	4	7	8
12	1	12	2	11	9	10	11	28	5	12

$R1 \bowtie R2$

Natural Join

A	x	B	y	w	D	z
1	12	2	11	5	6	30

```
29  SELECT * FROM r2 NATURAL JOIN r1;
```

Data Output Messages Notifications

	b integer	y integer	a integer	w integer	d integer	z integer	x integer
1	2	11	1	5	6	30	12

$R_1 \bowtie_{(R_1.A > R2.z \text{ OR } R1.A >= R2.w) \text{ and } R1.y = R2.y)} R2$

R1.A	R1.X	R1.B	R1.Y	R2.B	R2.W	R2.D	R2.Y	R2.A	R2.Z
7	2	6	11	2	5	6	11	1	30
10	7	2	4	4	7	8	4	7	9